

Notes: Autor, Dorn, and Hanson (AER, 2013)
**The China Syndrome: Local Labor Market Effects of Import Competition in the
United States**

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1 Big picture

- **Question.** What are the local labor-market effects of the rapid rise in Chinese import competition in the United States between 1990 and 2007?
- **Setting.** The paper studies 722 mainland U.S. commuting zones (CZs). CZs differ in their exposure to China because they had different initial industry specializations before the import shock arrived.
- **Core shock.** U.S. goods imports from China rose extremely quickly: from about \$26 billion in 1991/1992 to \$330 billion in 2007, while U.S. exports to China remained much smaller. Over roughly the same period, U.S. manufacturing employment as a share of the working-age population fell sharply.
- **Main empirical object.** A shift-share measure of Chinese import exposure per worker: CZs receive more exposure when they initially employ many workers in industries where national U.S. imports from China grow rapidly.
- **Core identification problem.** U.S. imports from China may rise because China becomes more productive, but they may also rise because U.S. industry demand changes. The paper wants the China-supply component, not U.S.-demand shocks.
- **Instrument.** The authors instrument U.S. import growth from China using Chinese import growth in eight other high-income countries: Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland. The idea is that if China becomes more competitive in an industry, exports to many rich countries should rise.
- **Main manufacturing result.** In the preferred specification, a \$1,000 increase in Chinese import exposure per worker over a decade reduces the manufacturing employment-to-working-age-population ratio by about **0.60 percentage points**.
- **Aggregate benchmark.** Using total observed import growth, China exposure can explain about **44%** of the 1990–2007 decline in the U.S. manufacturing employment share. A more conservative supply-shock-only calculation gives about **21%**.
- **Beyond manufacturing.** The losses are not absorbed by migration or reemployment elsewhere. Import-exposed CZs experience lower overall employment, higher unemployment, higher labor-force nonparticipation, lower wages, and higher transfer payments.
- **Key local-labor-market result.** Manufacturing jobs disappear, but workers do not rapidly move to other CZs. Instead, more adjustment occurs within the local labor market through nonemployment, lower nonmanufacturing wages, and public transfers.
- **Distributional message.** Trade may generate aggregate gains through lower prices, variety, and inputs, but the local adjustment costs are concentrated and persistent. The paper is mainly about the distributional and medium-run adjustment consequences of globalization.
- **Economic takeaway.** The China shock is not just an industry-level manufacturing shock. It is a geographically concentrated labor-demand shock that spills over to the whole local economy.

2 Data and measurement (key objects)

2.1 Unit of observation: commuting zones

- **Local labor markets.** The paper uses commuting zones (CZs), which are clusters of counties connected by commuting flows.
- **Coverage.** The analysis includes **722 mainland U.S. CZs**, covering both metropolitan and rural areas.
- **Why CZs?** They are meant to approximate local labor markets better than states or counties. A CZ is large enough to contain many commuting relationships but small enough to capture local exposure to industry shocks.
- **Why local markets matter.** If workers are not perfectly mobile, and if local wages and employment do not instantly equilibrate nationally, then import competition can have persistent local effects.

2.2 Trade data

- **Import data.** The paper uses UN Comtrade data on U.S. imports from China at the six-digit Harmonized System product level.
- **Sample start.** 1991 is the first year with sufficiently comparable product-level trade data across many high-income countries.
- **Main periods.** The empirical work mainly compares 1990–2000 and 2000–2007, treating pooled ACS data from 2006–2008 as 2007.
- **China trade growth.**
 - U.S. imports from China: \$26.3 billion in 1991/1992 to \$330.0 billion in 2007.
 - U.S. exports to China: \$10.3 billion in 1991/1992 to \$57.4 billion in 2007.
 - The central asymmetry is that import growth is much larger than export growth.
- **Comparison countries.** The same trade information is constructed for eight other developed economies used in the IV strategy.

2.3 Industry employment and local exposure

- **Industry employment.** CZ-by-industry employment comes from County Business Patterns.
- **Industry detail.** The paper uses employment in **397 manufacturing industries**.
- **Exposure variation.** CZs differ in exposure because they initially specialize in different manufacturing industries.
- **Not just manufacturing share.** The initial manufacturing share explains less than one-quarter of the variation in import exposure. Much of the useful variation comes from *which* manufacturing industries are located in a CZ.

2.4 Labor-market outcomes

- **Population, employment, unemployment, wages, and income.** These are built from Census IPUMS for 1970, 1980, 1990, and 2000, plus pooled ACS data for 2006–2008.
- **Main population.** Most outcomes are for working-age individuals, ages 16–64.
- **Employment categories.** The paper separates manufacturing employment, nonmanufacturing employment, unemployment, and not-in-labor-force status.
- **Wages.** The wage outcome is mean log weekly earnings, so it combines hourly wages and hours worked.
- **Household income.** The paper studies household income per working-age adult, including wage and salary income, business and investment income, and transfer income.

2.5 Transfer-payment outcomes

- **Sources.** Public transfer data come from the BEA Regional Economic Accounts and the Social Security Administration.
- **Programs.** The paper studies total individual transfers and components including:
 - Trade Adjustment Assistance (TAA),
 - unemployment benefits,
 - Social Security retirement,
 - Social Security Disability Insurance (SSDI),
 - Medicare and Medicaid,
 - SSI, TANF/AFDC, and SNAP,
 - education and training assistance.
- **Important measurement point.** TAA is the transfer program explicitly targeted to trade-displaced workers, but it turns out to be a tiny part of the total fiscal response.

2.6 Key descriptive facts

- **Import exposure rose faster in the 2000s.** Median ten-year-equivalent exposure growth rose from \$0.89 thousand per worker in 1990–2000 to \$2.11 thousand in 2000–2007.
- **Large geographic variation.** In 2000–2007, the 75th percentile CZ had exposure growth of \$3.11 thousand per worker, versus \$1.60 thousand at the 25th percentile.
- **Highly exposed large CZs.** San Jose and Providence are among the most exposed large CZs in both periods.
- **Manufacturing share declined.** The manufacturing employment share of the working-age population fell from 12.69 in 1990 to 10.51 in 2000 and 8.51 in 2007.
- **Transfers rose substantially.** Average individual transfers per capita rose from \$3,338 in 1990 to \$5,544 in 2007.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theory motivation: why local industry mix matters

The theory section motivates the empirical design with a model in which regions differ in their industry composition. China becomes more productive or faces lower trade costs in some industries. U.S. regions that initially employ many workers in those industries are more exposed.

A simplified version of the traded-sector employment implication is:

$$\hat{L}_{Ti} \approx -\tilde{\alpha} \sum_j \frac{L_{ij}}{L_{Uj}} \frac{M_{jU}^C \hat{A}_j^C}{L_{Ti}}. \quad (1)$$

Interpretation.

- i indexes regions and j indexes industries.
- L_{ij}/L_{Uj} is region i 's share of national employment in industry j .
- M_{jU}^C is U.S. imports from China in industry j .
- $\hat{A}_j^C$ is the China supply/productivity shock in industry j .
- Regions with more initial employment in industries where China becomes more competitive suffer larger reductions in traded-sector labor demand.

The model also clarifies the role of trade imbalance. With balanced trade, import competition in some regions could be offset by export demand in others. With large U.S. trade deficits vis-à-vis China, the import shock can shrink U.S. traded-sector employment in the aggregate, not just reallocate jobs across regions.

3.2 Main import-exposure measure

The paper's main exposure variable is the change in Chinese import exposure per worker in CZ i during period t :

$$\Delta IPW_{it}^u = \sum_j \frac{L_{ijt}}{L_{ujt}} \frac{\Delta M_{cjt}^u}{L_{it}}. \quad (2)$$

Interpretation.

- ΔM_{cjt}^u is the change in U.S. imports from China in industry j .
- L_{ijt}/L_{ujt} allocates national industry-level imports to CZs according to each CZ's initial share of national employment in that industry.
- L_{it} scales exposure by the CZ's start-of-period employment.
- This is a shift-share measure: the *shift* is national industry import growth from China; the *share* is the CZ's initial industry employment mix.

The key identifying variation is not whether a region is manufacturing-intensive in general, but whether it is specialized in industries where Chinese import growth is large.

3.3 Instrumental variable

The instrument replaces U.S. import growth from China with Chinese import growth in other high-income countries and uses lagged employment shares:

$$\Delta IPW_{it}^o = \sum_j \frac{L_{ij,t-1}}{L_{uj,t-1}} \frac{\Delta M_{cjt}^o}{L_{i,t-1}}. \quad (3)$$

Interpretation.

- ΔM_{cjt}^o is the change in Chinese exports to other high-income countries in industry j .
- The comparison countries are Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland.
- The instrument is strong if Chinese export growth to other rich countries predicts Chinese export growth to the United States in the same industries.
- Lagging employment shares helps reduce simultaneity bias: if current employment already responds to anticipated China competition, using prior-decade shares is safer.

3.4 Main 2SLS manufacturing regression

The core estimating equation is:

$$\Delta L_{it}^m = \gamma_t + \beta_1 \Delta IPW_{it}^u + X_{it}' \beta_2 + e_{it}. \quad (4)$$

Interpretation.

- ΔL_{it}^m is the ten-year-equivalent change in manufacturing employment as a share of the working-age population.
- γ_t is a period dummy for 2000–2007 when the two periods are stacked.
- X_{it} contains start-of-period controls.
- ΔIPW_{it}^u is instrumented by ΔIPW_{it}^o .
- The coefficient β_1 answers: how much does the local manufacturing employment share change when Chinese import exposure rises by \$1,000 per worker?

Because the regression is in first differences, it removes time-invariant CZ differences. In the stacked specification, the model is close to a three-period fixed-effects design but allows a less restrictive error structure.

3.5 Control variables

The preferred specification includes:

- start-of-period manufacturing employment share,
- Census division dummies,

- share of population that is college educated,
- share foreign born,
- female employment rate,
- share of employment in routine-intensive occupations,
- average occupational offshorability.

Interpretation. These controls address the concern that China exposure proxies for preexisting manufacturing decline, regional trends, immigration patterns, changing female employment, routine-biased technological change, or offshoring of tasks rather than import competition.

3.6 Beyond-manufacturing outcomes

The same 2SLS design is then applied to broader local outcomes:

$$\Delta Y_{it} = \gamma_t + \beta \Delta IPW_{it}^u + X_{it}' \theta + \varepsilon_{it}, \quad (5)$$

where Y_{it} is alternatively:

- log working-age population,
- manufacturing employment,
- nonmanufacturing employment,
- unemployment,
- labor-force nonparticipation,
- SSDI receipt,
- mean log weekly wages,
- government transfers,
- household income.

Interpretation. This section asks whether the China shock is absorbed by migration, offset by nonmanufacturing job growth, capitalized into wages, or shifted onto government transfers.

3.7 Alternative trade-exposure measures

The paper checks whether the results depend on the exact definition of exposure.

Domestic plus international exposure. This adds the effect of China competing with U.S. producers in third-country markets:

$$\sum_j \frac{E_{ijt}}{E_{ujt}} \frac{\Delta M_{cjt}^u + \sum_{o \neq c} \left(X_{ojt}^u / X_{ojt} \right) \Delta M_{cjt}^o}{E_{it}}. \quad (6)$$

Net imports. This subtracts U.S. exports to China from U.S. imports from China:

$$\sum_j \frac{E_{ijt}}{E_{ujt}} \frac{\Delta M_{cjt}^u}{E_{it}} - \sum_j \frac{E_{ijt}}{E_{ujt}} \frac{\Delta X_{ujt}^c}{E_{it}}. \quad (7)$$

Factor content of net imports. This converts net imports into embodied labor services:

$$\sum_j \frac{E_{ijt}}{E_{ujt}} \frac{\tilde{E}_{uj0}}{V_{uj0}} \frac{\Delta M_{cjt}^u}{E_{it}} - \sum_j \frac{E_{ijt}}{E_{ujt}} \frac{\tilde{E}_{uj0}}{V_{uj0}} \frac{\Delta X_{ujt}^c}{E_{it}}. \quad (8)$$

Interpretation. These alternatives ask whether the baseline result is really about gross imports, or whether it survives once the paper accounts for third-market competition, imported intermediate inputs, U.S. exports to China, gravity-based China productivity shocks, and the factor content of trade.

3.8 Welfare calculations

The paper does not estimate the full gains from trade. Instead, it calculates two medium-run efficiency costs of adjustment: distortionary transfer financing and involuntary labor-force nonparticipation.

The starting condition for labor supply is:

$$w_0 u_c(y + w_0 h_0, h_0) = -u_h(y + w_0 h_0, h_0). \quad (9)$$

This says that at the initial optimum, the utility value of the consumption financed by an extra hour of work equals the marginal disutility of working that hour.

Worker surplus from employment is approximated as:

$$\Delta = w_0 h_0 - \frac{w_0 h_0}{1 + 1/\eta_h} = \frac{w_0 h_0}{\eta_h + 1}. \quad (10)$$

After a wage reduction, the deadweight loss from involuntary employment loss is approximated by:

$$\Delta' = \frac{\alpha w_0 h_0 [1 + \eta_h (\alpha - 1)]}{\eta_h + 1}, \quad \alpha = \frac{w'_0}{w_0}. \quad (11)$$

Interpretation.

- The paper distinguishes voluntary labor-supply responses to lower wages from involuntary job loss.
- It uses labor-supply elasticities from the public-finance literature to infer how much of the observed employment decline is involuntary.
- These calculations are not the total welfare effect of China trade. They are partial measures of adjustment frictions.

4 Identification and instruments (what makes the estimates credible)

4.1 Source of identifying variation

- The empirical design compares CZs that had different initial industry structures.

- A CZ more specialized in furniture, apparel, toys, electronics, or other China-exposed industries receives a larger predicted shock than a CZ specialized in less-exposed industries.
- The design is not simply comparing manufacturing regions with nonmanufacturing regions. The preferred specification controls directly for initial manufacturing share.

4.2 Why the IV matters

OLS may be biased toward zero for two reasons:

- **Demand shocks.** If U.S. demand rises in an industry, both U.S. imports from China and U.S. employment may rise, masking the negative causal effect of import competition.
- **Measurement error.** CZ-by-industry employment shares are imperfect, especially at detailed industry levels.

The IV attempts to isolate the China supply component: Chinese productivity growth, reductions in trade costs, policy reforms, and WTO accession that increase China’s export capacity across rich-country destinations.

4.3 First-stage strength

- Figure 2 shows a strong first stage: a \$1,000 predicted increase in exposure per worker corresponds to about a \$815 increase in measured exposure.
- In Table 3, the preferred first-stage coefficient is **0.631** with a standard error of **0.087**.
- The first stage remains strong after adding initial manufacturing share, Census division dummies, demographic controls, routine-task controls, and offshorability controls.

4.4 Pretrend and falsification evidence

- Table 2 regresses earlier manufacturing employment changes from 1970–1990 on future China exposure from 1990–2007.
- The pooled pre-exposure coefficient is positive and statistically insignificant, not a large negative pretrend.
- This helps rule out the simple story that places later exposed to China were already on a long-run relative decline before China became a major exporter.

4.5 Controls for competing explanations

- The paper controls for routine-task intensity to separate import competition from computerization or routine-biased technological change.
- It controls for occupational offshorability to distinguish goods import competition from task offshoring.
- It includes Census division dummies to absorb broad regional trends.
- It controls for initial demographics, immigration, and female employment.

4.6 Robustness to alternative trade measures

- Results are similar when exposure includes Chinese competition in foreign markets.
- Results are similar when imports are net of intermediate inputs.
- Results are similar when the exposure measure subtracts U.S. exports to China.
- Results are similar under a gravity-residual approach that isolates changes in China’s relative productivity and market access.
- Results are similar when exposure is measured by the factor content of net imports.

4.7 Main limitations

- **Correlated demand shocks.** The IV assumes that common Chinese export growth to the United States and other rich countries is mainly a China supply shock, not a rich-country demand shock common across destinations.
- **U.S. productivity shocks.** If weak U.S. productivity in an industry causes both U.S. employment declines and Chinese import growth in many rich-country markets, the IV may still be contaminated.
- **Local exposure is a proxy.** Imports are not physically shipped to the exposed CZs. Exposure is measured by industry employment structure, not actual local import consumption.
- **Aggregate effects require extrapolation.** The regression identifies relative local effects. Translating them into aggregate U.S. manufacturing job losses requires additional assumptions.
- **Full welfare is not estimated.** The paper measures some adjustment costs, but not consumer gains, variety gains, firm input gains, or long-run productivity effects.

5 Tables and figures (how to read the evidence)

5.1 Figure 1: China import penetration and U.S. manufacturing employment

Read: The figure plots two aggregate time series: the ratio of U.S. imports from China to total U.S. goods expenditure and the share of the working-age population employed in manufacturing.

- Chinese import penetration rises sharply, especially after 2001.
- Manufacturing employment per working-age population declines over the same period.
- The figure motivates the question but does not identify causality.

Why it matters. The aggregate pattern shows why China is a plausible candidate explanation for part of the U.S. manufacturing decline, but the paper’s causal evidence comes from cross-CZ variation and the IV strategy.

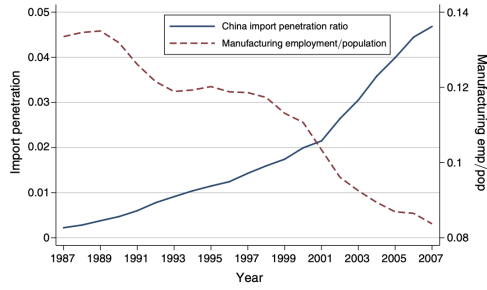


FIGURE 1. IMPORT PENETRATION RATIO FOR US IMPORTS FROM CHINA (left scale), AND SHARE OF US WORKING-AGE POPULATION EMPLOYED IN MANUFACTURING (right scale)

TABLE 1—VALUE OF TRADE WITH CHINA FOR THE US AND OTHER SELECTED HIGH-INCOME COUNTRIES AND VALUE OF IMPORTS FROM ALL OTHER SOURCE COUNTRIES, 1991/1992–2007

	I. Trade with China (in billions 2007 US\$)		II. Imports from other countries (in billions 2007 US\$)		
	Imports from China (1)	Exports to China (2)	Imports from other low-inc. (3)	Imports from Mexico/ CAFTA (4)	Imports from rest of world (5)
<i>Panel A. United States</i>					
1991/1992	26.3	10.3	7.7	38.5	322.4
2000	121.6	23.0	22.8	151.6	650.0
2007	330.0	57.4	45.4	183.0	763.1
Growth 1991–2007	1,156%	456%	491%	375%	137%
<i>Panel B. Eight other developed countries</i>					
1991/1992	28.2	26.6	9.2	2.8	723.6
2000	94.3	68.2	13.7	5.3	822.6
2007	262.8	196.9	31.0	11.6	1329.8
Growth 1991–2007	832%	639%	236%	316%	84%

Notes: Trade data is reported for the years 1991, 2000, and 2007, except for exports to China which are first available in 1992. The set of “other developed countries” in panel B comprises Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland. Column 3 covers imports from all countries that have been classified as low income by the World Bank in 1989, except for China. Column 4 covers imports from Mexico and the Central American and Caribbean countries covered by the CAFTA-DR. Column 5 covers imports from all other countries (primarily from developed countries).

5.2 Table 1: trade with China and other source countries

Read: Table 1 compares U.S. trade with China, trade with other low-income countries, trade with Mexico/CAFTA, and trade with the rest of the world.

- U.S. imports from China grow by **1,156%** from 1991/1992 to 2007.
- U.S. exports to China grow by **456%**, but from a much smaller base.
- Other rich countries also experience large increases in imports from China, which supports the instrument’s relevance.

Why it matters. China is not just one source country among many. It accounts for a very large share of the increase in U.S. imports from low-wage countries.

5.3 Appendix Tables 1 and 2: exposure variation and baseline CZ outcomes

Read: Appendix Table 1 describes the distribution of Chinese import exposure per worker across CZs. Appendix Table 2 describes baseline levels and changes in labor-market and transfer variables.

- In 2000–2007, ten-year-equivalent import exposure is \$3.11 thousand at the 75th percentile and \$1.60 thousand at the 25th percentile.
- This interquartile difference is economically meaningful: using the preferred coefficient, it implies a large differential manufacturing employment decline.
- Appendix Table 2 shows that manufacturing employment was falling while nonmanufacturing employment and transfers were rising.

Why it matters. The paper has enough cross-CZ variation to estimate local effects, and the magnitude of that variation is large enough to matter economically.

5.4 Figure 2: first stage and reduced form

Read: Figure 2 contains two added-variable plots.

- Panel A: the other-country import-growth instrument strongly predicts U.S. China import exposure. A \$1,000 predicted increase corresponds to about \$815 measured exposure.

APPENDIX

A. TABLES

APPENDIX TABLE 1—DESCRIPTIVE STATISTICS FOR GROWTH OF IMPORTS EXPOSURE PER WORKER ACROSS CZs: TEN-YEAR EQUIVALENT CHANGES

I. 1990–2000				II. 2000–2007			
<i>Panel A. Percentiles</i>							
90th percentile		2.05	90th percentile	4.30			
75th percentile		1.32	75th percentile	3.11			
50th percentile		0.89	50th percentile	2.11			
25th percentile		0.62	25th percentile	1.60			
10th percentile		0.38	10th percentile	1.03			
<i>Rank</i>							
<i>Panel B. Largest and smallest values among the 40 largest CZs</i>							
1	San Jose, CA	3.15	San Jose, CA	7.32			
2	Providence, RI	2.59	Providence, RI	4.99			
3	Buffalo, NY	2.24	Los Angeles, CA	3.59			
4	Boston, MA	1.55	San Diego, CA	3.08			
5	Portland, OR	1.53	Portland, OR	2.96			
6	San Diego, CA	1.52	Pittsburgh, PA	2.95			
7	Newark, NJ	1.32	Chicago, IL	2.93			
8	Los Angeles, CA	1.28	Milwaukee, WI	2.93			
9	Bridgeport, CT	1.27	Boston, MA	2.79			
10	Denver, CO	1.23	Dallas, TX	2.77			
20	Forth Worth, TX	0.83	Columbus, OH	1.90			
21	Phoenix, AZ	0.83	Phoenix, AZ	1.90			
31	Atlanta, GA	0.61	Fresno, CA	1.56			
32	Pittsburgh, PA	0.56	St. Louis, MO	1.53			
33	Sacramento, CA	0.53	Tampa, FL	1.49			
34	Kansas City, MO	0.51	Atlanta, GA	1.31			
35	West Palm Beach, FL	0.48	Baltimore, MD	1.25			
36	Fresno, CA	0.47	West Palm Beach, FL	1.22			
37	Orlando, FL	0.46	Kansas City, MO	1.13			
38	Houston, TX	0.45	Washington, DC	0.86			
39	Washington, DC	0.21	New Orleans, LA	0.70			
40	New Orleans, LA	0.19	Orlando, FL	0.59			

Notes: The table reports ten-year equivalent values of (Δ imports from China to US)/worker in kUS\$. The statistics in panel A are based on 722 CZs and weighted by start-of-period population size. The ranking in panel B is based on the 40 CZs with largest population in 1990, and indicates the largest city of each ranked CZ.

APPENDIX TABLE 2—MEANS AND STANDARD DEVIATIONS OF CZ LEVEL VARIABLES

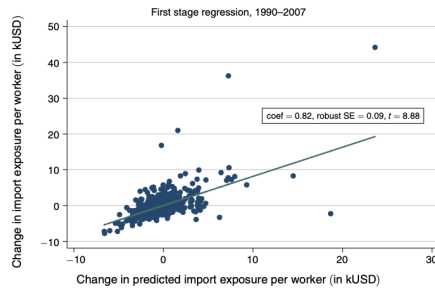
	I. Levels			II. Ten-year equivalent Δ s	
	1990/1991 (1)	2000 (2)	2007 (3)	1990–2000 (4)	2000–2007 (5)
(Imports from China to US)/ (workers in 1990) (in kUS\$)	0.29 (0.32)	1.32 (1.18)	3.58 (2.84)	1.14 (0.99)	n/a
(Imports from China to US)/ (workers in 2000) (in kUS\$)	0.25 (0.27)	1.08 (0.90)	2.92 (2.13)	n/a (2.01)	2.63 (2.01)
Percentage of working-age pop employed in manufacturing	12.69 (4.80)	10.51 (4.45)	8.51 (3.60)	–2.07 (1.63)	–2.73 (1.80)
Percentage of working-age pop employed in nonmanufacturing	57.75 (5.91)	59.16 (5.24)	61.87 (4.95)	1.29 (2.38)	3.70 (2.71)
Percentage of working-age pop unemployed	4.80 (0.99)	4.28 (0.93)	4.87 (0.90)	–0.51 (0.73)	0.85 (1.39)
Percentage of working-age pop not in the labor force	24.76 (4.34)	26.05 (4.39)	24.75 (3.70)	1.29 (2.56)	–1.82 (2.57)
Percentage of working-age pop receiving disability benefits	1.86 (0.63)	2.75 (1.04)	3.57 (1.41)	0.91 (0.63)	1.23 (0.71)
Average log weekly wage, manufacturing sector (in log pts)	655 (17)	666 (17)	671 (19)	11.4 (6.4)	7.8 (7.7)
Average log weekly wage, nonmanufacturing sectors (in log pts)	637 (16)	650 (15)	653 (16)	12.5 (4.1)	3.5 (4.3)
Average individual transfers per capita (in US\$)	3,338 (692)	4,297 (908)	5,544 (1,091)	1,004.4 (334.0)	1,844.0 (437.6)
Average retirement benefits per capita (in US\$)	1,121 (284)	1,262 (310)	1,398 (338)	150.5 (79.3)	206.2 (120.4)
Average disability benefits per capita (in US\$)	136 (46)	213 (77)	300 (112)	78.2 (39.8)	128.3 (61.5)
Average medical benefits per capita (in US\$)	1,115 (371)	1,789 (552)	2,564 (679)	698.3 (231.9)	1,142.8 (288.5)
Average federal income assistance per capita (in US\$)	298 (136)	270 (134)	303 (129)	–24.8 (43.6)	52.2 (46.0)
Average unemployment benefits per capita (in US\$)	106 (52)	86 (43)	108 (55)	–19.1 (29.4)	34.1 (41.0)
Average TAA benefits per capita (in US\$)	0.6 (0.6)	1.1 (1.0)	2.2 (2.7)	0.5 (0.9)	1.6 (3.3)
Avg household income per working-age adult (in US\$)	32,122 (6,544)	38,126 (7,743)	37,909 (7,501)	5,964 (2,358)	–367 (2,646)
Avg household wage and salary income per working-age adult (in US\$)	23,496 (4,700)	27,655 (5,449)	28,872 (6,304)	4,152 (1,569)	1,703 (2,623)

Notes: $N = 722$ CZs. Statistics in columns 1 and 4 are weighted by 1990 population, statistics in columns 2 and 5 are weighted by 2000 population, and statistics in column 3 are weighted by 2007 population. The first two rows of column 1 report import volumes for the year 1991, all other variables in column 1 are based on 1990 data. Information on employment composition, wages, and income in column 3 is derived from pooled 2006–2008 ACS data.

- Panel B: the reduced form shows that CZs with higher predicted exposure experience larger manufacturing employment declines.

Why it matters. The causal chain appears in the raw residualized data: other-country China exposure predicts U.S. China exposure, and predicted exposure predicts employment decline.

Panel A. 2SLS first stage regression, full sample



Panel B. OLS reduced form regression, full sample

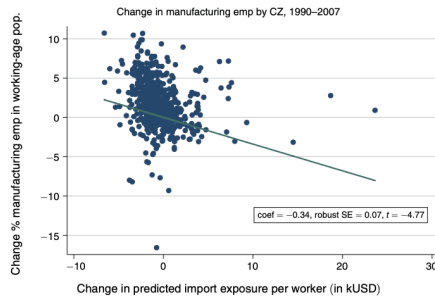


FIGURE 2. CHANGE IN IMPORT EXPOSURE PER WORKER AND DECLINE OF MANUFACTURING EMPLOYMENT: ADDED VARIABLE PLOTS OF FIRST STAGE AND REDUCED FORM ESTIMATES

Notes: $N = 722$. The added variable plots control for the start of period share of employment in manufacturing industries. Regression models are weighted by start of period CZ share of national population.

TABLE 2—IMPORTS FROM CHINA AND CHANGE OF MANUFACTURING EMPLOYMENT IN CZs, 1970–2007: 2SLS ESTIMATES
Dependent variable: $10 \times \text{annual change in manufacturing emp/working-age pop (in \% pts)}$

	I. 1990–2007			II. 1970–1990 (pre-exposure)		
	1990–2000 (1)	2000–2007 (2)	1990–2007 (3)	1970–1980 (4)	1980–1990 (5)	1970–1990 (6)
(Δ current period imports from China to US)/worker	–0.89*** (0.18)	–0.72*** (0.06)	–0.75*** (0.07)			
(Δ future period imports from China to US)/worker				0.43*** (0.15)	–0.13 (0.13)	0.15 (0.09)

Notes: $N = 722$, except $N = 1,444$ in stacked first difference models of columns 3 and 6. The variable “future period imports” is defined as the average of the growth of a CZ’s import exposure during the periods 1990–2000 and 2000–2007. All regressions include a constant and the models in columns 3 and 6 include a time dummy. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

5.5 Table 2: initial 2SLS estimates and pre-exposure placebo

Read: Table 2 estimates the basic effect of import exposure on manufacturing employment.

- 1990–2000 coefficient: **-0.89**.
- 2000–2007 coefficient: **-0.72**.
- Stacked 1990–2007 coefficient: **-0.75**.
- Interpretation: a \$1,000 increase in import exposure per worker reduces manufacturing employment per working-age population by about three-quarters of a percentage point.

Pre-exposure test.

- Future China exposure does not robustly predict manufacturing employment decline in 1970–1990.
- This weakens the concern that future China-exposed places were already on a uniquely declining path.

5.6 Table 3: preferred manufacturing employment estimates

Read: Table 3 adds controls step by step.

- Column 1 baseline stacked coefficient: **-0.746**.
- Column 2 controls for initial manufacturing share: **-0.610**.
- Column 6 preferred full-control coefficient: **-0.596**.
- Preferred first-stage coefficient: **0.631**.

Interpretation. Even after controlling for manufacturing share, regional trends, demographics, routine-task intensity, and offshorability, Chinese import exposure strongly reduces local manufacturing employment.

Economic magnitude.

- The preferred coefficient implies that a \$1,000 increase in exposure over a decade reduces manufacturing employment per working-age population by 0.596 percentage points.
- Total observed China import growth implies that China exposure can explain about 44% of the full 1990–2007 manufacturing employment decline.
- The more conservative supply-shock-only calculation explains about 21% of the full decline.

5.7 Table 4: population adjustment

Read: Table 4 asks whether workers leave exposed CZs.

- With full controls, the effect on total working-age population is essentially zero: **-0.050** with a large standard error.
- Effects are also not robust for college adults, noncollege adults, or age groups.

TABLE 3—IMPORTS FROM CHINA AND CHANGE OF MANUFACTURING EMPLOYMENT
IN CZs, 1990–2007: 2SLS ESTIMATES
Dependent variable: 10 × annual change in manufacturing emp/working-age pop (in % pts)

	I. 1990–2007 stacked first differences					
	(1)	(2)	(3)	(4)	(5)	(6)
(Δ imports from China to US)/ worker	−0.746*** (0.068)	−0.610*** (0.094)	−0.538*** (0.091)	−0.508*** (0.081)	−0.562*** (0.096)	−0.596*** (0.099)
Percentage of employment in manufacturing _{−1}		−0.035 (0.022)	−0.052*** (0.020)	−0.061*** (0.017)	−0.056*** (0.016)	−0.040*** (0.013)
Percentage of college-educated population _{−1}				−0.008 (0.016)		0.013 (0.012)
Percentage of foreign-born population _{−1}				−0.007 (0.008)		0.030*** (0.011)
Percentage of employment among women _{−1}				−0.054** (0.025)		−0.006 (0.024)
Percentage of employment in routine occupations _{−1}					−0.230*** (0.063)	−0.245*** (0.064)
Average offshorability index of occupations _{−1}					0.244 (0.252)	−0.059 (0.237)
Census division dummies	No	No	Yes	Yes	Yes	Yes
II. 2SLS first stage estimates						
(Δ imports from China to OTHI)/ worker	0.792*** (0.079)	0.664*** (0.086)	0.652*** (0.090)	0.635*** (0.090)	0.638*** (0.087)	0.631*** (0.087)
R ²	0.54	0.57	0.58	0.58	0.58	0.58

Notes: N = 1,444 (722 commuting zones × 2 time periods). All regressions include a constant and a dummy for the 2000–2007 period. First stage estimates in panel II also include the control variables that are indicated in the corresponding columns of panel I. Routine occupations are defined such that they account for 1/3 of US employment in 1980. The offshorability index variable is standardized to mean of 0 and standard deviation of 10 in 1980. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 4—IMPORTS FROM CHINA AND CHANGE OF WORKING-AGE POPULATION
IN CZ, 1990–2007: 2SLS ESTIMATES
Dependent variables: Ten-year equivalent changes in log population counts (in log pts)

	I. By education level			II. By age group		
	All (1)	College (2)	Noncollege (3)	Age 16–34 (4)	Age 35–49 (5)	Age 50–64 (6)
Panel A. No census division dummies or other controls						
(Δ imports from China to US)/worker	−1.031** (0.503)	−0.360 (0.660)	−1.097** (0.488)	−1.299 (0.826)	−0.615 (0.572)	−1.127*** (0.422)
R ²	—	0.03	0.00	0.17	0.59	0.22
Panel B. Controlling for census division dummies						
(Δ imports from China to US)/worker	−0.355 (0.513)	0.147 (0.619)	−0.240 (0.519)	−0.408 (0.953)	−0.045 (0.474)	−0.549 (0.450)
R ²	0.36	0.29	0.45	0.42	0.68	0.46
Panel C. Full controls						
(Δ imports from China to US)/worker	−0.050 (0.746)	−0.026 (0.685)	−0.047 (0.823)	−0.138 (1.190)	0.367 (0.560)	−0.138 (0.651)
R ²	0.42	0.35	0.52	0.44	0.75	0.60

Notes: N = 1,444 (722 CZs × two time periods). All regressions include a constant and a dummy for the 2000–2007 period. Models in panel B and C also include census division dummies while panel C adds the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period commuting zone share of national population.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Interpretation. The local shock is not quickly arbitrated away by migration. This is crucial because if workers moved rapidly, local unemployment, nonemployment, and wage effects would be much smaller.

5.8 Table 5: employment status

Read: Table 5 decomposes the adjustment of the working-age population.

- Log counts: a \$1,000 exposure shock reduces manufacturing employment by **4.23%**, raises unemployment by **4.92%**, and raises nonparticipation by **2.06%**.
- Population shares: manufacturing employment falls by **0.596 percentage points**; nonmanufacturing employment falls by **0.178**; unemployment rises by **0.221**; labor-force nonparticipation rises by **0.553**.
- The total employment-population ratio falls by about **0.77 percentage points**.
- SSDI receipt rises by **0.076 percentage points**.

Education split.

- Noncollege adults experience much larger total employment losses than college adults.
- For noncollege adults, the nonmanufacturing employment share also falls substantially.
- For college adults, nonmanufacturing employment rises modestly, partially offsetting manufacturing losses.

Interpretation. Manufacturing job loss spills into the broader local economy. The main adjustment margin is not migration or full reemployment; it is unemployment and labor-force exit.

5.9 Table 6: wages

Read: Table 6 estimates effects on average log weekly wages.

TABLE 5—IMPORTS FROM CHINA AND EMPLOYMENT STATUS OF WORKING-AGE POPULATION
WITHIN CZs, 1990–2007: 2SLS ESTIMATES
Dependent variables: Ten-year equivalent changes in log population counts
and population shares by employment status

	Mfg emp (1)	Non-mfg emp (2)	Unemp (3)	NILF (4)	SSDI receipt (5)
<i>Panel A. 100 × log change in population counts</i> (Δ imports from China to US)/worker	−4.231*** (1.047)	−0.274 (0.651)	4.921*** (1.128)	2.058* (1.080)	1.466*** (0.557)
<i>Panel B. Change in population shares</i>					
<i>All education levels</i> (Δ imports from China to US)/worker	−0.596*** (0.099)	−0.178 (0.137)	0.221*** (0.058)	0.553*** (0.150)	0.076*** (0.028)
<i>College education</i> (Δ imports from China to US)/worker	−0.592*** (0.125)	0.168 (0.122)	0.119*** (0.039)	0.304*** (0.113)	—
<i>No college education</i> (Δ imports from China to US)/worker	−0.581*** (0.095)	−0.531*** (0.203)	0.282*** (0.085)	0.831*** (0.211)	—

Notes: $N = 1,444$ (722 CZs × two time periods). All statistics are based on working age individuals (age 16 to 64). The effect of import exposure on the overall employment/population ratio can be computed as the sum of the coefficients for manufacturing and nonmanufacturing employment; this effect is highly statistically significant ($p \leq 0.01$) in the full sample and in all reported subsamples. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

TABLE 6—IMPORTS FROM CHINA AND WAGE CHANGES
WITHIN CZs, 1990–2007: 2SLS ESTIMATES
Dependent variable: Ten-year equivalent change in average log weekly wage (in log pts)

	All workers (1)	Males (2)	Females (3)
<i>Panel A. All education levels</i> (Δ imports from China to US)/worker	−0.759*** (0.253)	−0.892*** (0.294)	−0.614*** (0.237)
R^2	0.56	0.44	0.69
<i>Panel B. College education</i> (Δ imports from China to US)/worker	−0.757** (0.308)	−0.991*** (0.374)	−0.525* (0.279)
R^2	0.52	0.39	0.63
<i>Panel C. No college education</i> (Δ imports from China to US)/worker	−0.814*** (0.236)	−0.703*** (0.250)	−1.116*** (0.278)
R^2	0.52	0.45	0.59

Notes: $N = 1,444$ (722 CZs × two time periods). All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

- All workers: **-0.759 log points**.
- Males: **-0.892**; females: **-0.614**.
- College workers: **-0.757**; noncollege workers: **-0.814**.

Interpretation. Even among workers who remain employed, wages fall in more exposed CZs. The estimates may understate true wage pressure if lower-wage workers are more likely to exit employment.

5.10 Table 7: manufacturing versus nonmanufacturing wages

Read: Table 7 separates employment and wage effects by sector.

- Employment falls sharply in manufacturing: **-4.231%** for all workers.
- Manufacturing wages do not fall significantly.
- Nonmanufacturing wages fall sharply: **-0.761 log points** for all workers.
- Nonmanufacturing wage declines are similar for college and noncollege workers.

Interpretation. The wage effect appears mainly outside manufacturing. A plausible mechanism is that the manufacturing shock reduces local demand for nontraded services while displaced workers increase labor supply to nonmanufacturing jobs.

5.11 Table 8: government transfer receipts

Read: Table 8 studies how public transfers respond.

- A \$1,000 import-exposure increase raises total individual transfers by **\$57.73 per capita**.
- TAA rises strongly in percentage terms but only by **\$0.23 per capita**.
- SSDI rises by **\$8.40 per capita**.

	I. Manufacturing sector			II. Nonmanufacturing		
	All workers (1)	College (2)	Noncollege (3)	All workers (4)	College (5)	Noncollege (6)
<i>Panel A. Log change in number of workers</i> (Δ imports from China to US)/worker	−4.231*** (1.047)	−3.992*** (1.181)	−4.493*** (1.243)	−0.274 (0.651)	0.291 (0.590)	−1.037 (0.764)
R^2	0.31	0.30	0.34	0.35	0.29	0.53
<i>Panel B. Change in average log wage</i> (Δ imports from China to US)/worker	0.150 (0.482)	0.458 (0.340)	−0.101 (0.369)	−0.761*** (0.260)	−0.743** (0.297)	−0.822*** (0.246)
R^2	0.22	0.21	0.33	0.60	0.54	0.51

Notes: $N = 1,444$ (722 CZs $\times$ two time periods). All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

TABLE 8—IMPORTS FROM CHINA AND CHANGE OF GOVERNMENT TRANSFER RECEIPTS
IN CZs, 1990–2007: 2SLS ESTIMATES
Dep vars: Ten-year equivalent log and dollar change of annual transfer receipts per capita (in log pts and US\$)

	Total individual transfers (1)	TAA benefits (2)	Unemployment benefits (3)	SSA retirement benefits (4)	SSA disability benefits (5)	Medical benefits (6)	Federal income assist (7)	Educ/training assist (8)
<i>Panel A. Log change of transfer receipts per capita</i> (Δ imports from China to US)/worker	1.01*** (0.33)	14.41* (7.59)	3.46* (1.87)	0.72* (0.38)	1.96*** (0.69)	0.54 (0.49)	3.04*** (0.96)	2.78** (1.32)
R^2	0.57	0.28	0.48	0.36	0.32	0.27	0.54	0.33
<i>Panel B. Dollar change of transfer receipts per capita</i> (Δ imports from China to US)/worker	57.73*** (18.41)	0.23 (0.17)	3.42 (2.26)	10.00* (5.45)	8.40*** (2.21)	18.27 (11.84)	7.20*** (2.35)	3.71*** (1.44)
R^2	0.75	0.28	0.41	0.47	0.63	0.66	0.53	0.37

Notes: $N = 1,444$ (722 CZs $\times$ two time periods), except $N = 1,436$ in column 2, panel A. Results for TAA benefits in column 2 are based on state-level data that is allocated to CZs in proportion to unemployment benefits. Unemployment benefits in column 3 include state benefits and federal unemployment benefits for civilian federal employees, railroad employees, and veterans. Medical benefits in column 6 consist mainly of Medicare and Medicaid. Federal income assistance in column 7 comprises the SSI, AFDC/TANF, and SNAP programs while education and training assistance in column 8 includes such benefits as interest payments on guaranteed student loans, Pell grants, and Job Corps benefits. The transfer categories displayed in columns 2 to 8 account for over 85 percent of total individual transfer receipts. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

- Medical benefits rise by **\$18.27 per capita**.
- Federal income assistance rises by **\$7.20 per capita**.
- Retirement benefits rise by **\$10.00 per capita**.

Interpretation. The fiscal response is large, but it mostly comes through broad social insurance and income-support programs, not the trade-specific TAA program.

5.12 Table 9: household income

Read: Table 9 studies household income per working-age adult.

- Average total household income falls by **1.48%**, or about **\$493**.
- Average wage and salary income falls by **2.14%**, or about **\$549**.
- Social Security plus AFDC income rises by **2.12%**, or about **\$17**.
- Median household wage and salary income falls by **2.32%**, or about **\$477**.

Interpretation. Transfers offset only a small fraction of lost earnings. The local household income effect is economically large and negative.

5.13 Table 10: alternative exposure definitions

Read: Table 10 repeats the main outcomes using alternative exposure measures.

- Baseline gross imports: manufacturing employment coefficient **-0.60**.
- Domestic plus international exposure: **-0.51**.
- Net of intermediate inputs: **-0.49**.
- Net imports: **-0.45**.
- Gravity residual: **-0.29**.
- Factor content of net imports: **-0.57**.

TABLE 9—IMPORTS FROM CHINA AND CHANGE IN HOUSEHOLD INCOME, 1990–2007: 2SLS ESTIMATES
Dependent variable: Ten-year equivalent percentage and real dollar change in average
and median annual household income per working-age adult (in %pts and US\$)

	Average HH income/adult by source				Median HH income/adult	
	Total	Wage- salary	Business invest	SocSec + AFDC	Total	Wage- salary
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Percent change</i> (Δ imports from China to US)/worker	–1.48*** (0.36)	–2.14*** (0.59)	–0.51 (0.74)	2.12*** (0.58)	–1.73*** (0.38)	–2.32*** (0.51)
R^2	0.69	0.43	0.76	0.52	0.53	0.52
<i>Panel B. Dollar change</i> (Δ imports from China to US)/worker	–492.6*** (160.4)	–549.3*** (169.4)	40.1 (116.7)	17.3*** (4.3)	–439.9*** (112.7)	–476.5*** (122.2)
R^2	0.63	0.40	0.72	0.51	0.49	0.48

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). Per capita household income is defined as the sum of individual incomes of all working-age household members (age 16–64), divided by the number of household members of that age group. Total income comprises wage and salary income; self-employment, business, and investment income; social security and welfare income; and income from other nonspecified sources. Social security and welfare income in column 4 includes social security retirement, disability, and supplementary income, aid to families with dependent children (AFDC), and general assistance. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

TABLE 10—ADDING EXPOSURE TO INDIRECT IMPORT COMPETITION
OR EXPOSURE TO NET IMPORTS, 1990–2007: 2SLS AND OLS ESTIMATES
Dependent variables: Ten-year equivalent changes of indicated variables

	I. Employment/pop		II. Log wages		III. Transfers, wage inc	
	Mfg (1)	Nonmfg (2)	Mfg (3)	Nonmfg (4)	log transfers (5)	Avg log HH wage inc (6)
<i>Panel A. Baseline results: Gross Chinese imports per worker</i> (2SLS) (Δ imports from China to US)/ worker	–0.60*** (0.10)	–0.18 (0.14)	0.15 (0.48)	–0.76*** (0.26)	1.01*** (0.33)	–2.14*** (0.59)
<i>Panel B. Domestic plus international exposure to Chinese exports</i> (2SLS) (Δ domestic + intn'l exposure to Chinese imports)/worker	–0.51*** (0.08)	–0.12 (0.12)	0.16 (0.42)	–0.60*** (0.23)	0.87*** (0.27)	–1.77*** (0.49)
<i>Panel C. Exposure to final goods and intermediate inputs</i> (2SLS) (Δ imports from China to US net of i med inputs)/worker	–0.49*** (0.12)	–0.01 (0.20)	0.71 (0.52)	–0.41 (0.37)	0.84** (0.36)	–1.23 (0.82)
<i>Panel D. Net Chinese imports per worker</i> (2SLS) (Δ net imports of US from China)/ worker	–0.45*** (0.10)	–0.09 (0.15)	0.46 (0.42)	–0.47* (0.27)	0.73** (0.35)	–1.39** (0.58)
<i>Panel E. Change in China-US productivity differential</i> (OLS gravity residual) (Δ comparative advantage China gravity residual)	–0.29*** (0.04)	–0.03 (0.08)	0.04 (0.28)	–0.26* (0.15)	0.53*** (0.14)	–0.78*** (0.25)
<i>Panel F. Factor content of net Chinese imports per worker</i> (2SLS) (Δ factor content of net imports from China)/worker	–0.57*** (0.10)	–0.12 (0.15)	0.59 (0.50)	–0.66** (0.26)	0.81** (0.36)	–1.70*** (0.54)

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). The estimates in panel A correspond to the main results of the preceding Tables 5, 7, 8, and 9. The mean (and standard deviation) of the trade exposure variables is 1.88 (1.75) in panel A; 2.28 (2.17) in panel B; 1.46 (1.48) in panel C; 1.58 (1.66) in panel D; 1.40 (1.79) in panel E; and 1.50 (1.48) in panel F. The first stage coefficient estimate is 0.61 (s.e. 0.07) for the models in panel B; 0.72 (0.09) for the final goods import instrument and –1.05 (0.25) for the intermediate inputs import instrument in panel C; 0.70 (0.10) for the import instrument and –0.52 (0.08) for the export instrument in panel D; and 0.72 (0.07) for the import instrument and –0.28 (0.06) for the export instrument in panel F. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Interpretation. The main conclusions do not depend on one narrow definition of import exposure. The estimated effects are smaller under some alternative scalings, but the signs and broad patterns remain stable.

5.14 Section VII: efficiency losses from adjustment frictions

Read: The paper estimates two partial medium-run deadweight losses.

- Transfer-induced fiscal deadweight loss: about **\$33 per capita** over 1990–2007, with a range of roughly \$22–\$44.
- Involuntary nonemployment deadweight loss: about **\$43 per capita** in 1990–2000 and **\$69 per capita** in 2000–2007; the full-period range is roughly \$87–\$137.

Interpretation. These numbers are not total losses from trade. They are adjustment-cost components that can offset part of the gains from trade in the medium run.

6 Short summary

- The paper estimates the local labor-market effects of rising Chinese import competition using cross-CZ variation in initial industry specialization.
- The key exposure measure is a shift-share variable: national industry-level import growth from China allocated to CZs by initial industry employment shares.
- The key instrument uses Chinese export growth to other high-income countries to isolate China supply shocks.
- A \$1,000 increase in import exposure per worker over a decade reduces the manufacturing employment share by about 0.60 percentage points in the preferred specification.

- The shock does not generate large population outflows, so adjustment remains local.
- Manufacturing employment falls, unemployment rises, labor-force participation falls, SSDI receipt rises, and nonmanufacturing wages fall.
- Government transfers increase substantially, but trade-specific TAA benefits account for only a very small share of the fiscal response.
- Household wage and salary income falls sharply, and transfers replace only a small part of the earnings loss.
- Robustness exercises using alternative trade measures give the same broad message.
- The paper’s main lesson is that trade shocks have concentrated local adjustment costs even when trade may generate aggregate gains.

7 Conclusion

7.1 Core conclusion

Autor, Dorn, and Hanson show that the rise of Chinese import competition had large, geographically concentrated effects on U.S. local labor markets. The most exposed CZs experienced substantial losses in manufacturing employment, and these losses spilled over into broader employment, wage, income, and transfer outcomes.

7.2 Economic intuition

The core mechanism is simple: local economies specialize. When China becomes much more competitive in certain industries, U.S. regions specialized in those industries suffer a negative labor-demand shock. Because workers do not immediately move away or fully reallocate into other jobs, the local shock persists and spreads through the local nonmanufacturing economy.

7.3 Takeaway

The paper changes the way economists study import competition. Instead of focusing only on national skill groups or manufacturing industries, it shows that trade shocks operate through local labor markets. The China shock reduced manufacturing employment, lowered local earnings, raised transfer payments, and generated meaningful medium-run adjustment costs concentrated in exposed communities.